Mathematical overview

On a finite connected graph, nonnegative integer heights whose neighboring values differ by at most 1 can extend fixed clues exactly when every two clue heights differ by at most their shortest graph distance. Assuming at least one clue, the least and greatest extensions are L(v)=max(0,max_c(a_c-d(v,c))) and U(v)=min_c(a_c+d(v,c)). When an extension exists, both are legal; without one the formulas miss a clue. Closing a row adds a shorter route; it can make old clues impossible. The distance is number of edges, regardless of the drawing's physical lengths.

For two legal landscapes with the same fixed clues, their pointwise minimum and maximum are legal. The fewest single-site unit changes between them equals the sum of absolute height changes. One can lower the first landscape to the pointwise minimum, then reverse a lowering path from the second to that minimum. To justify a lowering step, choose a highest current site that exceeds the target. A neighbor one level higher could not also be an excess site (maximality), and if already at its target it would force an illegal target jump. Hence lowering this chosen site is legal. Integer heights and fixed clues matter; moves change exactly one unit at one site.

Consequently every integer total between sum L and sum U occurs. A legal raising path from L to U visits all those totals one at a time. This does not say an arbitrary intermediate pointwise selection is legal. Children explore graph geometry (2-5), shortest transitions and their explanation (4-5), and fixed cube budgets (K-5 with adult reading). Concrete trials establish witnesses and counterexamples; the distance and legal-step arguments prove the general facts.

Entry route, preparation, launch, and pacing

Unpiloted. Paper size and coordinates are digitally checked; cube fit and marker handling have not been physically rehearsed. Page 3 is a genuine young-child concrete entry: counting at most 11 cubes and comparing adjacent tower heights. Page 1 needs small height comparisons and following connecting edges; grades 2-5 can investigate independently after launch. Page 2 needs differences, a recoverable move record, and readiness for a general shortest-route explanation. Adults may read and record without choosing children's moves.

Per pair: the three printed pages, twelve stacking cubes, seven movable height markers, two visibly different fixed-clue markers, pencil, and scrap paper. A spare seven-site row can be drawn on scrap; heights 0-4 suffice for the printed transition. Large graph stations are 21.9 mm or 24.0 mm square. The marker board's column spacing is 24.3 mm and level spacing 20.1 mm; a marker sits on a horizontal line, not inside a cell. Zero is a marked height even with no cube.

For eleven children at KK11 / 3333 / 445, use five kits: sixty cubes, thirty-five movable markers, ten clue markers, five pencils, and one adult at each table. The third older child checks neighboring jumps and rotates into the pair. Give page 3 to the youngest table; offer pages 1 and 3 to the middle table; the older table can begin anywhere. The clue markers for page 2 must stay at A=G=1 while changing white heights.

Common launch, 4 minutes: let children stack cubes. Show a non-task three-site path with heights 1,2,2. Change the middle to 3 and let the partner reject the 1-to-3 jump. Point out that graph adjacency is a drawn line, so in a ring the last and first also count as neighbors. The base pages already demonstrate tower-to-marker-to-number conversion; on an encore return, rehearse one ordinary marker placement rather than introduce a new code. Do not supply the distance formulas or navigation algorithm at launch.

Suggested return visits: 15-25 minutes for graph clues, 20-35 for shortest transitions, 15-25 for budgets. A first route can be page 3 followed by one row-versus-ring comparison. Leave the general connectedness question available to older children beyond the first hour. Save a constructed ring or route before resetting, and record exactly which investigations children used.

Solutions, hints, and extensions

1. The open row has A=0,D=3 three steps apart, forcing B=1,C=2. E may be 2,3,4: exactly three completions. After closure, A and D are just two steps apart through E, so a difference of 3 is impossible. With D=2 the closed ring has exactly (A,B,C,D,E)=(0,0,1,2,1),(0,1,1,2,1),(0,1,2,2,1). The last simultaneously maximizes B and C. Optional hint: walk the shorter way between clue squares; each edge permits at most one level of change. Extension: add a shortcut edge to a legal graph and decide which clues survive; derive envelopes with distances before using formulas.

2. Eight moves is optimal: B,C must decrease by 2 each and E,F increase by 2 each. One legal route starts by lowering C: (1,2,3,2,1,0,1), (1,2,2,2,1,0,1), (1,1,2,2,1,0,1), (1,1,1,2,1,0,1), (1,0,1,2,1,0,1), (1,0,1,2,1,1,1), (1,0,1,2,2,1,1), (1,0,1,2,2,2,1), (1,0,1,2,3,2,1). Each changes one white site by one and keeps jumps at most one. No pair with the same clues requires extra moves beyond the sum of absolute changes, by the theorem above. Optional hint: ask which decrease would leave a neighboring jump of 2; keep the intended finish visible. Extension: repeat on a small branched graph; the same theorem works without a straight row. Changing fixed clues during the route is a different task.

3. The gray towers count in every total, as the page says. Least ring (1,0,0,1,0,0), total 2. Greatest (1,2,2,1,2,2), total 10. Every free site is adjacent to a clue of height 1 and hence at most 2; every site is nonnegative. Witness for budget 5: (1,0,0,1,1,2); budget 8: (1,1,1,1,2,2). Budget 11 is impossible. A child who still counts only the white towers gets every total 2 less (0 to 8), so 10 fails too; accept that with its reason. All budgets 2 through 10 occur. Exact counts by budget are 1,4,6,8,11,8,6,4,1 (49 rings total), not a required student catalog. Optional hint: distribute one cube at a time and let a partner check every edge. Extension: choose new clues and predict a total range; ask whether all totals occur on an arbitrary connected graph with consistent clues.

Sources, distinction, and checks

Base Week 47 studies rows, pointwise extrema and a five-move all-lowering K-1 transition. Page 1 changes the geometry to cycles and shortest routes; page 2 uses a mixed raising/lowering transition and the general exact-distance question; page 3 imposes a total budget and investigates absence of missing totals. These are three investigations within one theme.

Original extensions to the base shortest-distance/Lipschitz mathematics; no downloaded activity copied. The Week 1 encore and Math Circle by the Bay preface, printed pp. viii-x (PDF pp. 9-11), motivate return visits, not these specific problems. student/verify.py exhaustively checks row/ring completions, breadth-first searches the 120 permitted seven-site rows for the printed shortest route, and enumerates all 49 budget rings. General explanations are supplied separately above. No physical or classroom testing is claimed.